

Mohammed Sabbah

AI Engineer | LLM & Backend Systems

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SUMMARY

AI engineer specializing in LLM systems and agentic AI, with production experience building multi-tool RAG pipelines, LangGraph agents, and full-stack ML systems. Contributed to AI backend infrastructure at FlyRank AI and completed end-to-end ML pipeline development at Areisto. Projects include a document-grounded study assistant, a correctness-critical usage-metering and billing engine, real-time face recognition at millisecond latency, and a cross-platform plant disease classifier. Strong across the full ML lifecycle, from model training and evaluation to deployment and API integrations.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, SQL
- **ML & Deep Learning:** TensorFlow/Keras, PyTorch, Scikit-learn, Hugging Face Transformers
- **LLM & Agentic AI:** LangChain, LangGraph (multi-tool agent routing, stateful graphs), RAG, vector search (Chroma, FAISS), prompt engineering, LLM fine-tuning
- **Web & API Development:** FastAPI, Django (Basics), REST APIs, Streamlit, Redis, Stripe API
- **Computer Vision:** YOLO, FaceNet, OpenCV, Neural Networks, Transfer Learning
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, FAISS, PostgreSQL
- **DevOps & Tools:** Git/GitHub, Docker (Basic), Jupyter/Colab

EXPERIENCE

Backend AI Engineering Intern | FlyRank AI — Jul 2026 – Sep 2026

- Built and shipped backend APIs that expose AI model outputs to product clients, improving integration reliability for core AI features.
- Contributed core components to data ingestion and processing pipelines that feed ML models, enabling real-time AI workloads in production.
- Completed the Backend AI Engineering and AI Fluency tracks; capstone project (MeterFlow, a usage-metering and billing engine) accepted on final program review.

Machine Learning Trainee | Areisto Software Company — Gaza, Palestine — Apr 2025 – Jul 2025

- Completed a rigorous 2.5-month ML program, progressing from raw data preprocessing to full model deployment across supervised and unsupervised production tasks.
- Built end-to-end ML pipelines — data preprocessing, model training, evaluation, and deployment — delivering functional prototypes for each team project on schedule.
- Collaborated with a team of trainees on project-based work using Git, code reviews, and agile-style sprint cycles.
- Applied cross-validation and hyperparameter tuning across multiple models to improve accuracy and reduce overfitting on validation sets.

PROJECTS

MeterFlow – Usage Metering & Billing Engine (FastAPI · PostgreSQL · Redis · Stripe · React/TypeScript)

- Built a multi-tenant usage-metering backend with a two-layer idempotency design — Redis cache plus a database uniqueness constraint — guaranteeing exactly-once usage recording under concurrent retries.
- Implemented a two-stage quota gate — a fail-fast pre-check before invoking the metered AI provider, followed by an authoritative check against real token counts — enforcing exact plan-boundary behavior without wasting provider calls.
- Converted weighted token usage into a proportional plan-price charge, pricing cached-input, reasoning, and output tokens separately — verified correct (40% utilization on a \$29 plan yields \$11.60).
- Integrated Stripe subscription billing end-to-end in test mode: Checkout Session creation, signature-verified webhook handling, and automatic Free/Pro plan synchronization.
- Audited the existing codebase and fixed a tenant-isolation vulnerability, confirmed by automated tests: cross-tenant requests that previously returned 200 OK now correctly return 403 Forbidden.
- Built the full-stack React/TypeScript frontend (typed API client, TanStack Query, protected routing) consuming the FastAPI backend, including a live Stripe Checkout flow with webhook-verified subscription state.

StudyMate – AI Study Assistant (LangChain • LangGraph • RAG • Streamlit)

- Built a document-grounded AI study assistant letting users chat with, quiz, flashcard, and summarize uploaded PDFs, Word, PowerPoint, and text files within a single tool.
- Designed a LangGraph agent routing between chat/Q&A, quiz, flashcard, and summary tools based on document size and user intent, eliminating hard-coded single-pipeline logic.
- Implemented a size-aware RAG pipeline: small documents pass in full context; larger ones are chunked, embedded, and indexed in Chroma for retrieval-based Q&A.
- Engineered a batch-and-interleave generation strategy — chunk batching, round-robin interleaving, two-stage de-duplication — guaranteeing full-document coverage on large files.
- Built a word-level similarity de-duplication algorithm to filter near-duplicate questions without character-level false positives.
- Architected a services layer (DocumentManager, per-session contextvars, generic caching) to decouple tools from global state and safely isolate concurrent users.
- Shipped a 4-tab Streamlit UI (Chat, Quiz, Flashcards, Summary) plus a lightweight CLI, both backed by the same tools and agent core.

Quiz AI Generator Chatbot (LangChain • LangGraph • Streamlit)

- Built an AI chatbot generating interactive quizzes from PDF documents and natural language prompts using LangChain, LangGraph, and Mistral.
- Implemented agent memory and session persistence via LangGraph checkpointing and encrypted cookie-based user identification.
- Developed a modular tool system for automatic discovery and execution of LangChain tools, later extended into StudyMate's broader agent architecture.

Planto – Plant Disease Detection Mobile App (Flutter • TensorFlow)

- Built a cross-platform mobile app classifying plant diseases from leaf images in real time across multiple plant types.
- Fine-tuned and validated separate CNN sub-models per plant type, iterating on architecture and data augmentation to improve classification performance.
- Handled the full data pipeline and mobile deployment for both a general model and specialized per-plant sub-models.

Real-Time Face Recognition System (YOLO • FaceNet • FAISS • PostgreSQL)

- Built a real-time face recognition pipeline for live video streams using Python and OpenCV.
- Integrated YOLOv8 for face detection, FaceNet for embedding generation, and FAISS for millisecond-latency similarity search.
- Stored embeddings and user metadata in PostgreSQL, enabling scalable identity management across large datasets.

Deep Learning Architecture Research (ResNet • U-Net • VGG • AlexNet)

- Implemented and benchmarked four state-of-the-art architectures on image classification and segmentation tasks.
- Conducted systematic hyperparameter optimization and comparative analysis across custom and standard benchmark datasets.

EDUCATION

B.Sc. Computer Engineering | Islamic University of Gaza, Palestine — Sep 2020 – Jul 2025

GPA: 85.93% (3.5 / 4.0)

CERTIFICATIONS

- DeepLearning.AI TensorFlow Developer — Coursera
- TensorFlow: Data and Deployment — Coursera
- Neural Networks and Deep Learning — Coursera
- Introduction to Computer Vision and Image Processing — Coursera
- Machine Learning Specialization — Coursera
- Advanced English Language — Cambridge Training College Britain

LANGUAGES

- Arabic: Native Speaker
- English: Advanced